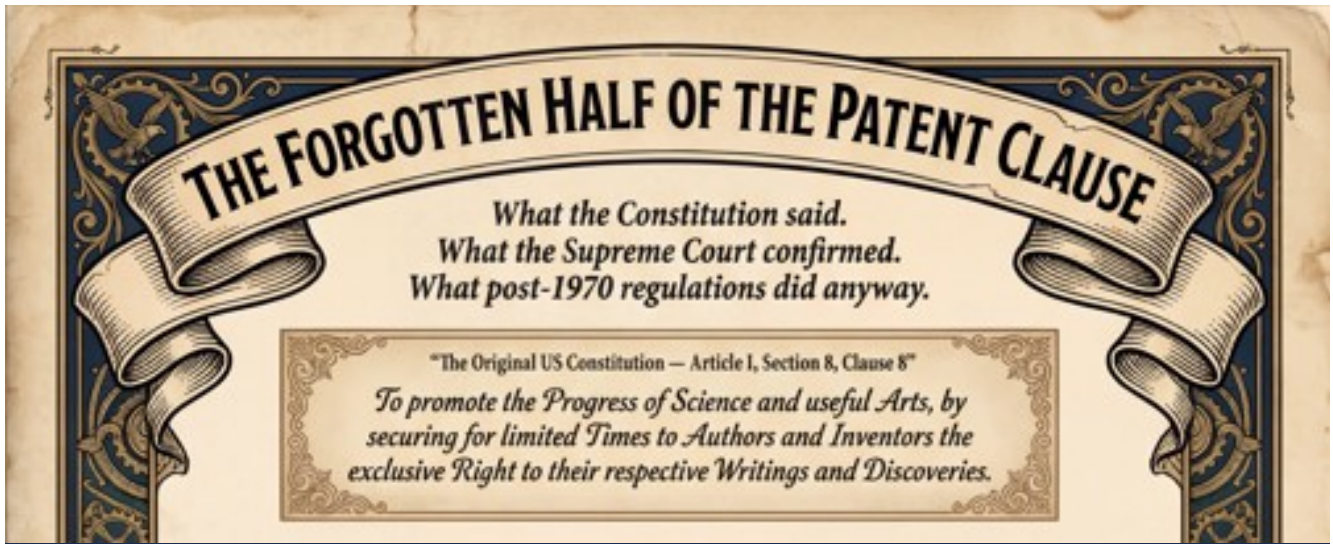




The Forgotten Half of the Patent Clause

*How Federal Regulations Doubled the Cost of American Living
and Rewrote the Constitution*

How a Constitutional Bargain Was Cancelled — Restoring It Frees Buyers from Surveillance,
Stops Robots Replacing America's Drivers & Operators, and Returns the Joy of Operating Real
Machines

A constitutional law litigation strategy and a citizen's playbook — in one book.

Raleigh Martin

Founder, the Make American Living Affordable (MALA) movement on X

This is the complete back-matter apparatus to the book above — its Endnotes, Endnote Commentary,
Supreme Court case notes, Glossary, and Key Terms, published online so that every source link is live
and the whole section is searchable.

Endnotes, Glossary & Key Terms

The complete back-matter apparatus, online

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Apparatus Convention

The book uses no superscript reference numbers in the running text. Citations of sources, court cases, and statutes appear in the Endnotes section at the back of the book, alphabetized by author surname (or, for cases, by case name). The Endnotes section is followed by an Endnote Commentary section, which collects in alphabetical order — keyed to the same author surname or case name as the Endnotes section — substantive notes on those sources: full passages a chapter quotes only in part, additional context the running text omits for the sake of pace, related material the curious reader will want and the impatient reader can skip. Not every Endnote has a corresponding Commentary entry; only those that genuinely reward a longer note. An Endnote that does have a Commentary entry ends with the italicized notice “*See Endnote Commentary.*” The two sections are designed to be read together by a reader who wants depth, and skipped together by a reader who does not. The Endnote Commentary is followed by a Glossary section, which collects in alphabetical order by abbreviation the acronyms used throughout the book — primarily federal agencies, statutes, and regulatory frameworks defined parenthetically at first use in the running text. The Glossary is followed by a Key Terms section, which collects in alphabetical order by term the book’s specific terms of art (such as bootleggers-and-Baptists, regulatory ratchet, anti-backsliding provision, three-tier breakdown, public domain in the patent-law sense, the Expired Patent Commons, remanufacturing in the

book’s specific sense, grandfather exemption, expired-patent commerce, constitutional forgetting, and two suspenders), each with a brief definition. The Endnotes section is formatted according to the Chicago Manual of Style, 17th edition. Each Endnote entry concludes with a URL — Justia for U.S. Supreme Court cases, Cornell University Legal Information Institute for federal statutes, Founders Online for Founding-era documents, the publisher’s canonical URL for periodicals, books, and reports, with Amazon as a last resort for older works without stable publisher URLs — so that readers using a digital edition of the book can follow the source directly.

The reader who wants to verify a claim in the running text consults the Endnote entry alphabetized by the author or case name as they appear in the text. There are no chapter-specific endnote groupings; the Endnotes section is a single book-wide alphabetical list. The reader who wants additional context on a source — the full passage a chapter quotes only in part, or material the running text omits for the sake of pace — consults the parallel entry in the Endnote Commentary section, which is keyed to the same anchor and ends with the italicized notice “*See Endnote for full citation.*” The reader who wants the meaning of an acronym consults the Glossary; the reader who wants to anchor a recurring concept consults the Key Terms section. Each section does one job.

In-text citations follow Chicago author-date style: integrated into the sentence as “Yandle (1983) named the pattern,” with full names and titles reserved for the Endnote entry. Acronyms are spelled out in full at first appearance in the book with the abbreviation following in parentheses; thereafter only the abbreviation is used.

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<https://www.amazon.com/dp/1107084334>. See *Endnote Commentary*.

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book; the methodologies closely parallel each other, the principal difference being that Metzl’s book presents an AI synthesis at its core while this book presents a human-discovered constitutional argument that the AI helped execute.

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Disease Control and Prevention. Analyzes FARS bicycle fatality data for the period 1975–2012 and reports cyclist mortality on a per-population rate basis (a methodology that addresses denominator concerns inherent in absolute-count comparisons). Cyclist mortality rate fell from 0.41 to 0.23 deaths per 100,000 across the study period — a 44 percent decline. The steepest decline occurred among children aged under 15; age-specific mortality rates for adults aged 35–74 increased over the same period. The bicycles section of the modern-bargain chapter cites the Vargo et al. rate figure as the population-adjusted counterpart to the chapter’s absolute-count IIHS figure, making possible an apples-to-apples comparison against the per-vehicle-mile fatality rate the chapter cites for the automobile.

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Wilkinson, David. U.S. Patent for a screw-cutting machine, granted 1798 (term expired 1812 under the fourteen-year rule of the Patent Act of 1790). The screw-cutting entry in the chapter's eight-product table. The Wilkinson design fell from a during-patent price of about four dollars per pound of cut screws to roughly sixty cents per pound by the 1880s, a sixty-four percent real-terms drop, as automated threading entered competing American shops.

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<https://en.zxauto.com.cn/CORPORATE.html>.

ZX Auto (Hebei Zhongxing Automobile Co., Ltd.). "GRANDTIGER." ZX Auto Global Official Site. Manufacturer product page for the ZX Grand Tiger pickup truck. Specifications include 2.4L gasoline and 2.8T diesel engine options, 5-speed manual transmission, 2WD/4WD, double cab, and a 5310×1812×1800mm body. The chapter on

patent-expired international vehicles cites the Grand Tiger as a working pickup truck derived from the 1997-to-2004 Toyota Hilux platform whose core mechanical patents have been in the public domain for more than a decade. Export pricing runs roughly \$13,000 to \$18,000 in the buyer markets the chapter discusses. ZX Auto, founded in 1949, is described on its own page as China’s first independent pickup brand.

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ZX Auto (Hebei Zhongxing Automobile Co., Ltd.). “TERRALORD.” ZX Auto Global Official Site. Manufacturer product page for the ZX Terralord pickup truck. Specifications include 4WD diesel powertrain, 6AT/6MT transmission, double cab, three trim levels (Standard, Luxury, Supreme Luxury), four-wheel disc brakes, and 265/65R17 tires. The chapter on patent-expired international vehicles cites the Terralord alongside the Grand Tiger as a working pickup truck derived from late-1990s Toyota Tacoma platform technology, exported by ZX Auto to more than ninety countries including Kenya, Ghana, Nigeria, the Philippines, Bolivia, and across Latin America.

<https://en.zxauto.com.cn/TERRALORD.html>.

Endnote Commentary

(Alphabetical by first significant element — author surname, or case name without “v.” — keyed to the corresponding entry in the Endnotes section above. Substantive notes only; not every Endnote has a Commentary entry.)

The Chicago Manual of Style. 17th ed. The book follows the 17th edition (2017) because it is the most widely used scholarly standard for U.S. nonfiction trade and academic publishing, and because most U.S. literary agents and academic-trade publishers expect endnote apparatus formatted according to its conventions. The 18th edition, published in 2024, refines a number of digital-citation conventions but does not alter the core bibliographic forms the book’s apparatus uses. Where this book’s conventions diverge from strict Chicago style, the divergence is intentional and noted. The principal divergence is that every Endnote entry concludes with a working URL pointing to the source itself — the

publisher's canonical page for periodicals, books, and reports; Justia for U.S. Supreme Court cases; the Cornell University Legal Information Institute for federal statutes; Founders Online for Founding-era documents — rather than to a citation database or aggregator. The aim is to put the reader in front of the authoritative location of each source. Some publisher canonical pages sit behind paywalls; the book does not provide workarounds for those paywalls, but it does name the canonical location so that any reader with institutional access, or any reader who chooses to purchase, can verify the citation directly at the source. Where the publisher's canonical URL is unavailable, the URL hierarchy falls back through DOI or scholarly database, with Amazon as a last resort for older works without stable publisher URLs. Court cases are linked to Justia. Federal statutes are linked to the Cornell University Legal Information Institute. Founding-era documents are linked to Founders Online at the National Archives.

Competitive-discount floor of fifteen percent. Two different fifteen-percent figures appear in the book, and they describe two different things. In the methodology chapter, the retrospective calculation of the \$15.73 trillion floor applied a competitive-pressure discount in a range of eight to fifteen percent — component two of that calculation, the price reduction felt by the buyers who still purchase a new item rather than a remanufactured one. That range is a minor adjustment riding atop the dominant component, the direct savings of roughly forty to sixty percent realized by the buyers who purchase remanufactured goods, conservatively drawn from an observed international spread of fifty to eighty-eight percent. The fifteen-percent figure used in the recovery chapters is a forward-looking estimate, not a restatement of that historical range: it is the conservative floor of the competitive discount a reopened commons would deliver after a favorable ruling. Three independent reasons make fifteen percent conservative rather than optimistic. First, in a post-ruling economy no durable good sits atop a fully patented bill of materials; somewhere in every supply chain remanufacturing introduces discounting, so the effect compounds through the chain rather than capping at the final assembly. Second, the roughly ninety countries where patent-expired remanufacturing is already lawful show real-dollar prices well below a fifteen-percent discount off the American equivalent — often a half or a third of the U.S. price — so fifteen percent understates what those markets actually

achieve. Third, the methodology’s “flowing” tier deliberately excludes a large mass of goods that would in fact be remanufactured once the fifty-eight percent blocked tier is freed, so the conservative base itself omits real savings. A wholly independent verification path points the same way: applying a twenty-five to forty percent savings rate (a thirty-four percent midpoint) to an \$81.1 trillion present-value base of consumer durable spending yields roughly \$28 trillion, comfortably above the \$15.73 trillion floor and confirming the floor is conservative from two directions. See the methodology chapter for the full calculation and the recovery chapters for the per-household figures.

Elster, Jon, ed. *Secrecy and Publicity in Votes and Debates*. The Italian parliament’s *voto segreto* practice operated under two distinct legal regimes across its 119-year history, a feature of the record the chapter’s compressed treatment cannot reveal in full. The complete timeline:

Period	Authority for voto segreto
1848 – 1922	Constitutional mandate under Article 63 of the Statuto Albertino: secret balloting was the constitutionally required method for every final vote on a law and for all votes of a personal nature. Seventy-four years on direct constitutional authority.
1922 – 1943	Fascist suspension. Mussolini’s March on Rome (October 1922) did not formally abolish the Statuto, but parliamentary practice was suspended in substance over the twenty-one years of Fascist rule.
1943 – 1948	Transition. Mussolini fell July 1943; the September 1943 armistice opened a five-year transitional period; a June 1946 referendum abolished the monarchy; a Constituent Assembly drafted a new Constitution.
1948 – 1988	Chamber rule under the Regolamento. The 1948 Republican Constitution formally abolished the Statuto Albertino in its entirety. Its Article 64 transferred voting procedure from constitutional authority to each chamber’s own rules, where voto segreto continued for the next forty years on chamber-rule authority alone.
1988 – present	Reformed. On October 13, 1988, the De Mita government repealed voto segreto as the default method for final votes through amendments to the Camera and Senato Regolamenti — no constitutional amendment required, because the practice no longer rested on constitutional ground. Voto segreto survives today for “conscience” votes (life-and-death, family, individual moral status) and personal-nature votes (individual members or ministers) only.

Article 63 of the Statuto Albertino, in English translation: *Votes shall be taken by rising, by division, and by secret ballot. The latter method, however, shall always be employed for the final vote on a law and in all cases of a personal nature.* That is a direct constitutional mandate, not a chamber preference. The 1848 framers of the Statuto thought secret balloting on final votes was important enough to put into the constitutional text itself, not merely into chamber rules.

The chapter’s argument that the United States could adopt Hybrid Voting by simple chamber-rule majority under Article I § 5 cl. 2 finds its closest historical analogue not in the Statuto-era mandate but in the post-1948 chamber-rule regime: forty years of working

secret-ballot practice, in a major Western parliamentary democracy, on chamber-rule authority alone, terminated when the chamber chose to terminate it. The American reform proposed in the Hybrid Voting chapter requires neither a constitutional amendment nor a chamber-rule mandate at the level the Statuto provided — only the simple-majority chamber-rule adoption the post-1948 Italian parliament already operated under for forty years. *See Endnote for full citation.*

Expired Patent Commons. This is the central term of art of this book, and the reason the book carries the title it does. When a patent reaches the end of its limited term, the invention it covered is meant to pass to the public — to be made, copied, improved, and remanufactured by anyone. The Supreme Court’s cases call that destination the public domain. This book calls it the Expired Patent Commons, and the choice of the word “commons” is deliberate. A commons is not an empty space or the mere absence of a right; it is a shared resource the public can actually reach and actually use, as villagers once walked onto common land and grazed it. An invention has entered the Expired Patent Commons only when an American may in fact pick it up and build it — the term is defined by use, not by paper. A patent that has lapsed on paper, but that federal regulation forbids anyone to manufacture, has not entered the commons; it is merely a patent that has expired. The Expired Patent Commons is the public’s half of Jefferson’s bargain — the forgotten half of the Patent Clause — and the book’s argument is that the post-1970 regulatory wall does not raid this commons but prevents it from ever forming. The term is defined in full in the chapter developing the Patent Clause prong.

Federal court procedure: party labels at each level. A reader who has followed the book through its chapters on the filable case and on plaintiff categories will at some point notice an apparent inconsistency in how parties to the Supreme Court’s cases are named. The chapters on filability and on who can sue use *plaintiff* and *defendant* throughout. The cases themselves — *Loper Bright*, *Cedar Point*, *Sheetz*, *Massachusetts v. EPA*, *Horne* — are captioned with the constitutional challenger sometimes first and sometimes second, and the Court’s opinions speak of *petitioner* and *respondent* rather than of plaintiff and defendant.

The inconsistency is not editorial. It is the way the federal court system labels the parties at each layer, and the labels shift as the case moves upward.

The terminology runs as follows. In federal district court, where every constitutional challenge to a federal regulation begins, the party who files the complaint — the grandfather-exemption advocate, in the case this book describes — is the *plaintiff*; the federal agency named as defendant (the EPA, NHTSA, or DOE, represented in court by the Department of Justice) is the *defendant*. The chapters of this book that describe who can file the case and what the complaint will ask for use those labels because that is where the case begins. At the United States Court of Appeals, whichever side lost in the district court and appealed becomes the *appellant*; the other side is the *appellee*. At the Supreme Court, the party who sought review by filing the petition for a writ of certiorari is the *petitioner*; the other side is the *respondent*. The case caption at the Supreme Court level always lists petitioner first.

Whether the grandfather-exemption advocate stands at the Supreme Court as petitioner or as respondent depends entirely on who won at the Court of Appeals. If the advocate loses at the circuit court and petitions the Supreme Court for review, the advocate is the *petitioner*. That is the procedural posture of every modern Supreme Court win this book builds on — *Loper Bright Enterprises v. Raimondo* (fishermen petitioned; won), *Cedar Point Nursery v. Hassid* (growers petitioned; won), *Sheetz v. County of El Dorado* (landowner petitioned; won), *Massachusetts v. EPA* (states petitioned; won), *Horne v. Department of Agriculture* (raisin growers petitioned; won). The constitutional challenger as petitioner is the modal posture in the cases the book relies on. If the advocate instead wins at the circuit court and the federal government petitions the Supreme Court, the advocate is the *respondent*. *Ruckelshaus v. Monsanto Co.* is the canonical example on the Takings Clause side: Monsanto had won below; the EPA Administrator petitioned; Monsanto stood as respondent and won again.

The plaintiff and defendant labels do not disappear when the case reaches the Supreme Court. They remain attached to the case’s history, and the briefs and the Court’s opinion will refer back to “the plaintiff below” or “the district-court defendant” when recounting the

procedural posture. But they are not the active labels at the Supreme Court. The advocates speak as petitioner and respondent. The caption reads “Petitioner v. Respondent.”

One usage note for the reader who reads journalism or general-interest writing on Supreme Court cases. Both routinely use *plaintiff* loosely to mean “the original challenger” all the way through the litigation, including at the Supreme Court. The usage is imprecise but common, and a reader who carries the imprecise usage into the Court’s own opinions will misread some of the citations. In a book written as a brief in waiting for an actual Supreme Court case, the precise terminology — plaintiff at the trial court, appellant or appellee at the circuit, petitioner or respondent at the Supreme Court — will read more credibly to the lawyer who eventually picks up the brief and works from it. *See Endnote for full citation.*

Ford Motor Company. “2025 Ford F-150 Lariat 4×4 SuperCrew 5.5’ Box.” The chapter’s title figure of \$72,495 is the out-the-door total a typical American family pays to drive a new F-150 home in 2025. It reflects the Lariat 4×4 SuperCrew 5.5’ Box — Ford’s mid-tier family trim, positioned above the work-truck XL (base \$37,250 MSRP) and the volume-trim XLT (approximately \$47,590 MSRP) but below the luxury-tier Platinum (\$74,905 base MSRP) and Raptor (\$79,005 base MSRP). The Lariat is what Ford positions as the family truck: the trim that combines hauling capacity with the interior appointments a household financing a new vehicle expects.

The components of the \$72,495 total reconstruct as follows. The Lariat 4×4 SuperCrew 5.5’ Box base MSRP for model year 2025 was \$65,255 at announcement, revised marginally during the production year as Ford adjusted incentives and option-package pricing. The destination and delivery charge added \$1,995, producing a manufacturer’s invoice subtotal of \$67,250. State and local sales tax, computed at approximately 7 percent on the manufacturer’s invoice subtotal — the U.S. weighted average across state and local jurisdictions in 2024–2025, per the Tax Foundation — added \$4,708. Title, license, and registration fees added approximately \$200, varying by state. The dealer documentation fee, which varies by state and dealership but commonly runs \$300 to \$700, added approximately \$337 at the median. The total at the dealer’s cashier — the amount the

buyer's lender finances or the buyer writes in a check — is \$72,490, within rounding of the \$72,495 the chapter uses.

The figure converges from a second verification path. TrueCar dealer-inventory listings show real-world 2025 Lariat 4×4 SuperCrew 5.5' Box PowerBoost Hybrid configurations at MSRP-with-destination totals of \$72,480, \$72,580, and \$72,690 — before tax and fees. The hybrid drivetrain adds approximately \$5,000 to the base Lariat's MSRP, but the verification is about the figure, not the engine: the base Lariat's out-the-door total and the hybrid Lariat's bare MSRP-with-destination converge at the same number, \$72,495, from two different directions. The convergence is what gives the chapter title figure its specificity.

The \$25,000 should-cost baseline against which the \$72,495 is compared is the book-canonical figure established in the wealth-transfer chapter. The \$47,495 per-vehicle wall premium is the arithmetic difference between what the family pays and what the family would pay for an equivalent-utility patent-expired vehicle absent the post-1970 federal regulatory architecture this book has been tracing. The reader who wants to reconstruct the out-the-door figure independently can do so from Ford's published 2025 MSRP schedule on shop.ford.com, the destination charge listed on every Ford window sticker, the reader's own state and local sales tax rate (or the Tax Foundation's national average), and a representative dealer documentation fee for the reader's state. *See Endnote for full citation.*

Jefferson, Thomas. “Letter to Isaac McPherson, August 13, 1813.” Chapter 2 quotes a single line from this letter — the line in which ideas spread “like fire, expansible over all space, without lessening their density in any point.” That line is one of three poetic images Jefferson stacks in a single passage, and it is not the most famous of the three. The full passage runs as follows:

John Deere W70. India product listing on IndiaMART. Source for the \$21,000–\$24,700 retail price range and the 2.5-acres-per-hour capacity figure for the 100-horsepower, 14-

foot, self-propelled wheeled combine harvester cited in the farm-equipment chapter as the India-market counterpart to the U.S. John Deere S7 600.

<https://dir.indiamart.com/items/john-deere-w70-14-feet-100-hp-combine-harvester-4-straw-walker-s22473.html>.

If nature has made any one thing less susceptible than all others of exclusive property, it is the action of the thinking power called an idea, which an individual may exclusively possess as long as he keeps it to himself; but the moment it is divulged, it forces itself into the possession of every one, and the receiver cannot dispossess himself of it. Its peculiar character, too, is that no one possesses the less, because every other possesses the whole of it. He who receives an idea from me, receives instruction himself without lessening mine; as he who lights his taper at mine, receives light without darkening me. That ideas should freely spread from one to another over the globe, for the moral and mutual instruction of man, and improvement of his condition, seems to have been peculiarly and benevolently designed by nature, when she made them, like fire, expansible over all space, without lessening their density in any point, and like the air in which we breathe, move, and have our physical being, incapable of confinement or exclusive appropriation.

Three images stand out. The first is the thinking-power image: an idea, once divulged, cannot be reclaimed. The second is the taper image — he who lights his taper at mine receives light without darkening me — which is the line most often quoted in patent scholarship and in economic writing on non-rival goods. The third is the fire-and-air image, which is the line Chapter 2 quotes. Each of the three has at one time or another been treated as the heart of the passage. The reader who wants Jefferson’s full argument — the part the chapter pivots to — will find the entire letter at the URL listed in the corresponding Endnote. *See Endnote for full citation.*

Martin, Roleigh. A note from the author, on the path that led to this book. The recognition that the post-1970 federal regulatory wall around expired-patent commerce constitutes a constitutional violation, rather than merely an unfortunate policy outcome, did not come to

me from a law degree or from a career in patent litigation. It came from two professional habits of mind I have spent a working life inside.

The first is game theory. My academic background trained me to look for the structure underneath an outcome — to ask, of any policy outcome whose stated rationale and observed effect do not match, what set of player incentives, what set of payoff matrices, and what set of moves the rules permit and forbid, would produce the observed effect from rational actors. Applying that lens to the post-1970 expansion of federal energy, safety, and emissions regulation produced, after some years of reading and re-reading the Patent Clause against the regulatory record, the recognition the book develops in detail: that the wall converts the patent bargain from a contract with two halves to a contract with one, and that the constitutional text already names the missing half. The half is in the words *limited Times*.

The second is the working habit of a man who spent thirty-four years in software — first as a programmer, then as a database programmer, inside the technical organization of a Fortune 25 company. The first joke I ever heard about my occupation, the one I have kept all the way through, is that *the worst programmer is the one who does exactly what you ask them, and nothing more*. The joke is funny because it is precisely right. A compiler does not interpret intent. It reads what you wrote, applies the language's rules, and returns either an executable result or a compilation error. It does not consult the spirit of the document; it consults the syntax of the document. A working life shaped by that machine produces a habit of reading any structured text as if it were code — including the Constitution.

Article I, Section 8, Clause 8, read by the compiler-reading mind, behaves like any other line of structured text. The phrase that arrests the reading is *limited Times*. It has a type signature. It is a duration. Durations have a start and an end. For a database programmer in particular, the same clause reads as a foreign-key constraint: the patent points at a defined term, the federal regulations that the rest of this book examines point at the patent-expired product, and the two tables join on the patent. When the patent's term expires, the parent row reaches end of life. Any regulation that continues to reference the patent-protected status as if it were live is holding a foreign key to a row that no longer exists in the form the

constraint requires. A database with referential integrity enabled refuses the operation. The federal government has been operating with referential integrity disabled since approximately 1970. The book is, in one of its readings, a fifty-year audit log of constitutional referential-integrity violations.

The two habits reinforce each other. Game theory asks what incentive structure would produce the wall as the equilibrium outcome of rational actors. Compiler-reading asks what the constitutional text, read as written, requires of the actors regardless of their incentives. The first habit explains why no one stopped the wall from being built. The second habit explains why the Constitution did not authorize what was built. The book is the long form of both observations.

The same compiler-reading habit, applied to the constitutional clause this book's Hybrid Voting chapter spends most of its constitutional argument on, returns a result the constitutional consensus has not. Article I, Section 5, Clause 3 reads in part: "the Yeas and Nays of the Members of either House on any question shall, at the Desire of one fifth of those Present, be entered on the Journal." Two phrases in that sentence admit ambiguities that two centuries of practice have papered over.

The first is *the Yeas and Nays of the Members*. Structurally a noun phrase, it could resolve to "the count of yeas and the count of nays" — a numeric tally — or to "the named position of each member" — a per-member roster. The text does not specify which. A compiler asked to resolve a noun phrase that admits two materially different parsings emits an error and demands clarification from a higher authority. The second is *on any question*. Structurally adverbial, it could resolve to "on every vote a chamber takes on the question" or to "on the question, once" — meaning a chamber that votes a bill twice could satisfy the clause by recording the named yeas and nays on either round. Again, the text does not specify. The constitutional consensus on both phrases is supplied by two centuries of legislative practice; legislative practice is not the type of authority a compiler accepts. A compiler accepts the text, the rules of the language, and binding constructions from a higher level of authority. Above the chamber, two such layers exist: the constitutional text itself, which on these two questions is ambiguous, and the Supreme Court's construction of the

text, which on these two questions does not exist. With both higher layers silent, the compiler accepts the next available authority — and the Constitution itself names what that authority is. Article I, Section 5, Clause 2 grants each chamber the power to *determine the Rules of its Proceedings*, and the rules of each chamber are, in practice, set by the procedural majority the chamber’s elected leadership commands. The Speaker of the House and the Majority Leader of the Senate, each elected by the chamber that operates under the Yeas-and-Nays clause, hold that procedural majority. The compiler-reading version of the political-question doctrine, then, is not an evasion the courts performed; it is a delegation the Constitution wrote, with two ambiguous phrases handed downward through the layers of authority until they come to rest with the leadership the chamber elected. The identity of that leadership is therefore not procedural housekeeping. It is the runtime variable on which the meaning of two constitutional ambiguities is set, and the chamber that elects its leader is, in the act of that election, choosing which resolution the Yeas-and-Nays clause will receive.

The Hybrid Voting chapter develops both of these textual ambiguities at greater length and closes with a line from Bernardo Bertolucci’s *The Last Emperor* — Reginald Johnston, played by Peter O’Toole, instructing the young Emperor Pu Yi: “If you cannot say what you mean, Your Majesty, you will never mean what you say. And a gentleman should always mean what he says.” The Patent Clause meant what it said, on both halves of the bargain it described. The Yeas-and-Nays clause, on these two questions, did not.

See Endnotes for full citations to the author’s articles and working paper.

Martin, Roleigh, ed. *ShareDebate International*. *ShareDebate International* was a small-press project at the leading edge of pre-Web electronic publishing. The magazine was a freeware diskette publication — which in 1990 meant a 1.44-megabyte 3.5-inch disk written in IBM PC format, copied by thousands of bulletin-board system operators, and read by their users one at a time on personal computers running DOS. After Issue 6 the publication moved from shareware to copyrighted freeware, eliminating the suggested subscription fee while preserving the distribution structure. Contributors across the magazine’s run included Milton Friedman, Murray Rothbard, George Gilder, Doug

Bandow, Eamonn Butler, Jerry Pournelle, Leonard Peikoff, Ben Bova, J. Neil Schulman, Victor Koman, Joseph Sobran, George Benston, George Kaufman, Lt. Gen. Daniel O. Graham, Paul Levinson, Douglas Ostrom of the Japan Economic Institute, Masanobu Taniguchi, the Libertarian Party, the Republican Liberty Caucus, and the League for Programming Freedom (with Richard Stallman). Ben Bova praised the magazine as “a bold, exciting venture” deserving of broad attention; J. Neil Schulman called it the arrival of “grassroots mass-market publishing.”

Two threads in the magazine bear directly on the present book. The first is Debate Topic #4, the “Japanese VS American Patent Law Debate,” opened by the editor in Issue 1 (1990) with the framing: *Is the Japanese Patent Law more just and more conducive to economic growth than America’s Patent Laws? A side aspect to this topic is the fairness and economic impact of America’s patent laws.* That side aspect — the fairness and economic impact of America’s patent laws — is a variant of the question this book takes up thirty-five years later in a deeper constitutional form. Issue 3 carried a four-page engagement letter from Nobel Laureate Milton Friedman, dated July 26, 1990, on Hoover Institution letterhead, praising the project’s premise, raising three substantive critiques, and wishing the editor well. Friedman returned to the magazine across multiple subsequent issues. The second relevant thread is Debate Topic #6, “Government’s Impact on Health Costs,” which prefigures the broader cost-of-living argument the author’s MALA (Making American Living Affordable) project has taken up in the years since. Third-party commentary on the magazine appears in Easton (2006), cited separately above. *See Endnote for full citation.*

Ostrom, Elinor. *Governing the Commons*. Elinor Ostrom was awarded the 2009 Nobel Prize in Economics for her work on the management of common-pool resources — the first woman to receive the prize, and one of the few political scientists ever to receive it. The award recognized a body of work spanning thirty years and hundreds of empirical cases: fisheries, irrigation systems, forests, grazing lands, groundwater basins. Her 1990 book *Governing the Commons* is the synthesis. Its central claim is that human communities, given the right institutional design, have repeatedly and successfully governed shared resources without resort to either state ownership or full privatization — the two solutions

Garrett Hardin’s “Tragedy of the Commons” had told mid-twentieth-century policy thinkers were the only options. Ostrom showed Hardin had been wrong about the empirical record. Real commons regimes, when they worked, depended on a small number of recurring design principles: clearly defined boundaries, rules matched to local conditions, collective-choice arrangements for those affected, monitoring, graduated sanctions, low-cost dispute resolution, and recognition by higher-level authorities. The reader who wants to see how those principles map onto the post-expiration patent commons described in this chapter — and why the post-1970 federal regulatory state has destroyed the higher-level recognition the commons depends on — will find the framework developed across the diagnostic and remedy chapters of the book. *See Endnote for full citation.*

Ranalli, Brent, James D’Angelo, and David King. *The Sunshine Reforms and the Transformation of Congressional Lobbying*. Brent Ranalli is a historian; James D’Angelo is a political scientist; David King is a senior lecturer at the Harvard Kennedy School. Their joint work at the Congressional Research Institute is the most rigorous body of empirical research available on the structural transformation of the U.S. Congress after the 1970 Legislative Reorganization Act. The 2019 *Foreign Affairs* essay “The Dark Side of Sunlight,” co-authored by D’Angelo and Ranalli, is the landmark synthesis; access to that essay sits behind a paywall. The 2018 Congressional Research Institute paper *The Sunshine Reforms and the Transformation of Congressional Lobbying* develops the same empirical case at greater length and is freely available on the Congressional Research Institute website (

<https://congressionalresearch.org>). The empirical record this book uses on lobbyist growth, on the explosion of legislation since 1970, and on the relationship between the “sunshine” transparency reforms and the rise of lobbyist surveillance derives from their joint research at that institute. Three findings are central. First, the registered-lobbyist count in Washington grew from a few hundred in 1970 to over thirteen thousand by 2024. Second, the volume of federal legislation tripled in the decade after the 1970 Act. Third, the federal government has not passed a budget that reduced the national debt in a single fiscal year since 1957. The reader who wants to understand why the modern American Congress produces the legislative outputs it does should begin with the Congressional Research

Institute’s published work. American constitutional self-government depends on citizens understanding what was changed in October of 1970 and what the change has cost. *See Endnote for full citation.*

Risner, Genevieve. “Parts of a Western Saddle and the Variations.” The Buffalo Bill Center of the West, located in Cody, Wyoming, is a Smithsonian Affiliate that operates five museums under one roof — the Buffalo Bill Museum, the Plains Indian Museum, the Whitney Western Art Museum, the Cody Firearms Museum, and the Draper Natural History Museum. Risner’s 2021 article on the parts and types of the Western saddle is the chapter’s recommended one-stop online reference for visual identification of the saddle types this chapter touches. The article identifies and photographs four distinct American saddle types from the Center’s own collection: the Vaquero saddle, the stock saddle, the McClellan, and the Pony Express saddle. Each is presented with photographs from multiple angles and a short narrative on the working role it served. The article’s evolution timeline runs from the eighth-century Moorish saddle, through the sixteenth-century Mexican vaquero adaptations that added the roping horn, through the 1850s adoption of the McClellan by the U.S. Military, through the 1860–1861 Pony Express. Although the article does not address the patent-expiration economics this chapter foregrounds, it is the standard accessible reference for the working forms the post-1797 commons produced. *See Endnote for full citation.*

Sierra Saddlery School. “Some History of the Western Saddle.” The chapter does not lean on this source for the dates and patent figures it cites — those come from the patent record itself and from the standard scholarly references. The Sierra Saddlery School blog post earns its place in the apparatus for a different reason: it is a competent and accessible narrative of how the working saddle of the American West came to look the way it does. The trans-Atlantic line runs from the Moorish horsemen of the Dark Ages, who carried into the Iberian peninsula a high-cantle saddle designed for armored war, to the Spaniards who adapted it into the Spanish War Saddle and brought it to the New World, to the post-conquest evolution of the saddle into a working tool of the Mexican vaquero. The vaqueros, in turn, were the saddle-makers who taught the trade to the working cowboy of Texas,

California, and the Great Basin, and the Texican-and-Californio split among working cowboys produced the two saddle silhouettes the modern reader still recognizes — the swell-fork, square-skirted, double-rigged Texican saddle suited to the heavy brush and “hard-and-fast” roping of the Texas range, and the slick-fork, round-skirted, center-fire-rigged Californio saddle suited to the milder climate, the dally style of roping, and the time the vaquero had for tooled-leather artistry. The reader who wants the longer saddle-history story than the chapter has room to tell will find a serviceable summary at the URL given in the corresponding Endnote, with the caveat that the post is a saddle-school blog rather than a museum or scholarly source. The Buffalo Bill Center of the West (Risner 2021) is the better one-stop reference for visual identification of the saddle types the chapter discusses. *See Endnote for full citation.*

Appendix A: General Notes on the Supreme Court Cases

(Alphabetical by case name. A brief, plain-language note on what each Supreme Court case was about, for the reader who meets the case in the text and wants the dispute behind it. Full citations and links are in the Endnotes section above. This section covers forty Supreme Court cases — twenty-one that affirm the patent bargain and “limited Times” on the merits, discussed in the chapters on the two constitutional prongs, and nineteen that do not by themselves speak to “limited Times” but are necessary to restore it in federal court, discussed in the chapters on filability, plaintiff categories, and objections considered. Each entry is labeled with its group in the year parenthetical. Chapter 1 reports the cumulative arsenal as “at least thirty-eight separate holdings”; the actual count here is forty, and the “at least” framing in the chapters is intentional, so that future expansions of this Appendix do not require revision of the chapter text — only of the Endnotes and of this section.)

Aetna Life Insurance Co. v. Haworth (1937; covered in the filability and objections-considered chapters). Aetna had issued disability-income policies to Edwin Haworth and had been paying benefits for several years when it concluded that Haworth was no longer disabled within the meaning of the contracts; Haworth disagreed but had not yet sued for past or future benefits. Aetna sought a federal-court declaration that the policies were void and no further benefits were owed. The Supreme Court held that the federal Declaratory Judgment Act of 1934 allowed Aetna to file: where a controversy is real and immediate, a

federal court may declare the parties' legal rights without waiting for one of them to be sued or prosecuted. The case is the foundation of the constitutional-litigation path this book describes — the path on which the patent-bargain case will be filed before any plaintiff has been arrested, fined, or enjoined.

Andrus v. Allard (1979; covered in the filability and objections-considered chapters). The federal Eagle Protection Act, joined by the Migratory Bird Treaty Act, made it unlawful to sell artifacts incorporating eagle parts even where the parts had been lawfully obtained before the statute. The owners of pre-existing inventories of Indian artifacts containing eagle feathers sued, claiming the sale ban was a Fifth Amendment taking of their property. The Supreme Court unanimously rejected the claim. A regulation that destroys one of the strands in the bundle of property rights — here, the right to sell — is not necessarily a taking, the Court reasoned, because possession, transport, and noncommercial transfer of the artifacts remained lawful. The case is the leading adverse Takings precedent the patent-bargain case will have to distinguish; the chapter on objections considered develops the distinction.

Arkansas Game & Fish Commission v. United States (2012; covered in the two constitutional prong chapters). The federal government, operating a dam, released water in a pattern that repeatedly flooded a state-owned wildlife area during the tree-growing season over a span of years, then stopped. The State sued, claiming the recurring flooding had killed timber and worked a taking. The government argued that flooding can be a taking only when it is permanent. The Supreme Court rejected that categorical rule. Whether government action is a taking, the Court held, depends on its actual effect on the property — not on whether the invasion is labeled temporary or permanent.

Aronson v. Quick Point Pencil Co. (1979; covered in the filability and objections-considered chapters). An inventor licensed an unpatented keyholder design to a manufacturer in exchange for a royalty, with a contractual provision reducing the royalty if no patent ever issued. No patent did. The manufacturer eventually argued that the royalty contract was unenforceable as inconsistent with federal patent policy, since the keyholder's design was free for all to copy. The Supreme Court rejected the argument. Federal patent

law leaves the unpatented public domain free for everyone, but it does not bar private parties from contracting on their own terms for the use of a design, and a state contract law that lets them do so is not preempted. The case is the modern restatement of the *Patterson v. Kentucky* principle that patent law and ordinary regulation coexist.

Bolling v. Sharpe (1954; covered in the filability and objections-considered chapters). The federal companion to *Brown v. Board of Education*, decided the same day. *Brown* held that state-mandated school segregation violated the Fourteenth Amendment’s Equal Protection Clause. *Bolling* faced the parallel question for the District of Columbia, which is not a state and to which the Fourteenth Amendment does not apply. The Court held that the Fifth Amendment’s Due Process Clause incorporates an equal-protection guarantee against the federal government. Federal action classifying similarly situated persons unequally is subject to constitutional review under the Fifth Amendment even where the Fourteenth Amendment has no purchase. The case is what makes any federal-equal-protection framing of the regulatory wall possible at all.

Bonito Boats, Inc. v. Thunder Craft Boats, Inc. (1989; covered in the two constitutional prong chapters). A Florida statute forbade using a “direct molding” process to duplicate unpatented boat hulls. *Bonito Boats* had designed a fiberglass hull it never patented; *Thunder Craft* was accused of copying it by the molding method the statute banned. The Supreme Court struck the statute down, holding that federal patent law leaves unpatented and patent-expired designs free for the public to copy, and that a State may not pull them back out of that public domain. The case is a cornerstone of the rule that what patent law does not protect, it affirmatively leaves open. Justice O’Connor grounded the holding in the Patent Clause itself, writing that “free exploitation of ideas will be the rule, to which the protection of a federal patent is the exception.”

Cedar Point Nursery v. Hassid (2021; covered in the two constitutional prong chapters). A California regulation gave labor organizers a right to enter an agricultural employer’s property to speak with workers for several hours a day during much of the year. The growers said the forced access took their property. The Supreme Court agreed. A regulation that appropriates a right to physically enter private land and hands it to others is a *per se*

taking, the Court held, even though the access was intermittent and the owners kept every other use of their land. The decision is part of the modern Court’s broadened reading of what counts as a taking.

Chevron U.S.A. Inc. v. Natural Resources Defense Council, Inc. (1984; covered in the two constitutional prong chapters). An environmental group challenged how the EPA had defined a term in the Clean Air Act. The Supreme Court ruled for the agency and, in doing so, announced a framework that governed administrative law for the next forty years: where a statute is ambiguous, a reviewing court should defer to any reasonable interpretation the administering agency adopts. “Chevron deference” became one of the most cited doctrines in American law — and the reason an agency’s reading of its own statute was rarely second-guessed. It was overruled in 2024 by *Loper Bright*.

City of Cleburne v. Cleburne Living Center, Inc. (1985; covered in the filability and objections-considered chapters). A Texas city denied a special-use permit to a proposed group home for persons with intellectual disabilities, citing zoning concerns the record could not substantiate. The Supreme Court struck the denial down under rational-basis review, finding that the only conceivable rationale for the decision was prejudice against an unpopular group, which is not a legitimate government interest. The case is the cleanest illustration of “rational basis with bite,” the narrow escape route from *Williamson v. Lee Optical* that allows an equal-protection challenge to succeed where identifiable animus, rather than a regulatory rationale, is what the court sees.

Dolan v. City of Tigard (1994; covered in the filability and objections-considered chapters). A hardware-store owner applied for a permit to expand her store and pave her parking lot. The city granted the permit on condition that she dedicate part of her property to a public greenway and bike path. She objected that the conditions were unrelated in scope to the impact of her expansion. The Supreme Court agreed with her on the second prong of the modern proportionality test it had begun building seven years earlier in *Nollan v. California Coastal Commission*: a regulatory condition imposed on the use of property must bear an essential nexus to the harm the regulation claims to address (*Nollan*) and must be roughly proportional in scope to that harm (*Dolan*). The two-prong test underlies *Sheetz v. County*

of El Dorado (2024), which extended it from individual administrative decisions to fees set by general legislation.

Eldred v. Ashcroft (2003; covered in the filability and objections-considered chapters). The Sonny Bono Copyright Term Extension Act of 1998 extended existing and future copyright terms by twenty years. Eric Eldred and others who relied on works passing into the public domain argued that retroactively extending existing terms violated the Patent and Copyright Clause’s “limited Times” requirement and the First Amendment. The Supreme Court, seven to two, upheld the Act. Justice Ginsburg’s majority held that “limited Times” requires only that the term be limited, not perpetual, and that Congress could rationally decide to extend the period of incentive even retroactively. Justice Breyer’s dissent argued the extension produced a term economically equivalent to ninety-nine point eight percent of a perpetual copyright. The case is the leading adverse Patent Clause precedent; the chapter on objections considered distinguishes it as a copyright case in a calendar-adjustment posture, not a patent case in a structural-ratchet posture.

First English Evangelical Lutheran Church of Glendale v. County of Los Angeles (1987; covered in the two constitutional prong chapters). A county ordinance temporarily barred a church from rebuilding on flood-prone land it owned. The church argued the ordinance had taken its property for the period the ban was in force. The Supreme Court used the case to settle what the Takings Clause requires: it does not forbid the government from taking property, but it requires the payment of just compensation when the government does. A finding that a regulation effects a taking establishes that compensation is owed; it is not, by itself, an order striking the regulation down.

Florida Prepaid Postsecondary Education Expense Board v. College Savings Bank (1999; covered in the two constitutional prong chapters). College Savings Bank sued a Florida state agency for infringing its patent. The question was whether Congress could strip the States of sovereign immunity and expose them to patent-infringement suits; the Court held that, in the manner Congress had attempted, it could not. The premise the Court worked from is what matters for this book, and the Court stated it plainly: a patent is a form of property, protected as other property is.

Graham v. John Deere Co. (1966; covered in the two constitutional prong chapters). A dispute over the validity of a patent on a clamp for plow shanks. The Supreme Court used the case to set out how courts decide whether an invention is “non-obvious” enough to deserve a patent — and, more important for this book, to describe the constitutional nature of the Patent Clause itself. The Court called the clause both a grant of power to Congress and a limitation on that power. In the Court’s own words, the clause “is both a grant of power and a limitation,” and “Congress in the exercise of the patent power may not overreach the restraints imposed by the stated constitutional purpose.”

Horne v. Department of Agriculture (2015; covered in the two constitutional prong chapters). A Depression-era federal marketing order required raisin growers to surrender a portion of each year’s crop to a government committee. The growers refused and were fined. The government argued the requirement was a permissible regulation of the raisin market, not a taking. The Supreme Court disagreed. Compelling the growers to hand over their raisins was a physical appropriation of property and therefore a taking per se, and the government could not avoid that conclusion by characterizing the appropriation as a condition of doing business or a contribution to orderly marketing.

Hunt v. Washington State Apple Advertising Commission (1977; covered in the filability and objections-considered chapters). North Carolina passed a statute requiring closed shipping containers of apples to display the U.S. grade rather than any state grade. The Washington State Apple Advertising Commission, representing Washington apple growers, sued, claiming the statute burdened interstate commerce. The Supreme Court used the case to set out the three-prong test for when a trade or membership association may sue on behalf of its members: at least one member must have standing in his own right, the interests at stake must be germane to the association’s purpose, and neither the claim nor the relief may require the participation of individual members. The case is the doctrinal basis for associational standing — the theory under which trucker associations, farm associations, and remanufacturer associations could sue in the patent-bargain case.

J. W. Hampton, Jr., & Co. v. United States (1928; covered in the chapter on the Patent Clause prong). An importer challenged a tariff the President had adjusted under authority

Congress granted him in the Tariff Act of 1922, arguing that Congress had handed away its own legislative power. The Supreme Court, in an opinion by Chief Justice Taft, upheld the delegation, holding that Congress may authorize another body to fill in the details of a statute so long as Congress supplies an intelligible principle to guide it. The case is the foundation of the modern administrative state and the authority for every agency that makes binding rules. The book cites it for that proposition and for its built-in limit: a delegated power can be no greater than the power Congress held to delegate. Congress could not delegate to an agency an authority Congress itself lacked — and Congress lacked any authority to cancel the public’s ownership of an invention whose patent has expired.

James v. Campbell (1882; covered in the two constitutional prong chapters). An inventor held patents on improvements used in machinery the federal government employed to print postage stamps and currency, and claimed the government’s use infringed them. The Supreme Court held that the government’s use of a patented invention is a taking of private property for which the inventor must be compensated — establishing that a patent is property in the full constitutional sense, not a privilege the government may draw on at will.

Kimble v. Marvel Entertainment, LLC (2015; covered in the two constitutional prong chapters). The inventor of a Spider-Man web-shooting toy licensed his patent to Marvel and kept collecting royalties after the patent expired. The Supreme Court held he could not: once a patent expires, royalties tied to it must stop, because the invention then belongs to the public. Though the Court chose to keep an older rule it admitted was imperfect, it restated the bedrock principle in plain words: when a patent expires, “the right to make or use the article, free from all restriction, passes to the public.”

Loper Bright Enterprises v. Raimondo (2024; covered in the two constitutional prong chapters). A group of commercial herring fishermen challenged a federal rule requiring them to pay the salaries of the monitors stationed on their boats. The Supreme Court used the case to overrule Chevron deference. A court, the Court held, must determine for itself what a statute means, using the ordinary tools of interpretation, rather than deferring to an

agency's reading. The decision reshaped administrative law, and it is what leaves a court free to decide the patent-bargain question on its own.

Loretto v. Teleprompter Manhattan CATV Corp. (1982; covered in the takings-clause prong chapter). A New York statute required landlords to permit cable-television operators to install a small cable box and a few feet of cable on the exterior of apartment buildings, in exchange for a one-time payment of one dollar set by a state commission. Jean Loretto, a New York City landlord, challenged the mandate as a taking. The Supreme Court held that any permanent physical occupation of property authorized by government, however minor, is a *per se* taking — categorically requiring compensation regardless of the economic impact on the owner or the public interest in the occupation. The case is the *per se* physical-occupation line that runs through *Horne v. Department of Agriculture* (2015) and *Cedar Point Nursery v. Hassid* (2021).

Lucas v. South Carolina Coastal Council (1992; covered in the two constitutional prong chapters). A developer bought two beachfront lots, after which a new state coastal statute barred him from building anything on them. The Supreme Court held that when a regulation deprives an owner of all economically beneficial use of his land, it is a taking without any case-by-case balancing — a taking *per se*. Total deprivation of use, the Court reasoned, is different in kind from the partial diminution of value that ordinary regulation produces, and it requires compensation categorically.

Massachusetts v. Environmental Protection Agency (2007; covered in the filability and objections-considered chapters). Twelve states and several local governments sued the EPA over the agency's refusal to regulate greenhouse-gas emissions from new motor vehicles. The Supreme Court held that Massachusetts had standing to sue, because the state had suffered concrete injury to its territory (rising sea levels eroding coastline) and to the welfare of its citizens, and because the federal government's failure to regulate was traceable to that harm and redressable by a court order. The decision announced the doctrine of *special solicitude*: a sovereign plaintiff is owed a more permissive standing analysis than the showing required of an ordinary private plaintiff. The case is the inflection

point for sovereign and quasi-sovereign standing in modern federal litigation, and the door through which states will be among the first plaintiffs in the patent-bargain case.

McCormick Harvesting Machine Co. v. Aultman (1898; covered in the two constitutional prong chapters). After a patent had been issued, its owner returned it to the Patent Office to have certain claims reissued; the Office instead treated the original patent as cancelled. The Supreme Court held the Patent Office had no such power. Once a patent issues, it becomes the property of the patentee, and only a court — not the executive agency that granted it — may set it aside. The case fixes the separation-of-powers line this chapter relies on.

MedImmune, Inc. v. Genentech, Inc. (2007; covered in the filability and objections-considered chapters). MedImmune licensed a patent from Genentech and continued paying royalties even as it doubted the patent’s validity. It sued for a declaratory judgment that the patent was invalid, while continuing to pay so that Genentech could not terminate the license and sue for infringement. The federal courts had been holding that a licensee who continued paying could not meet the case-or-controversy requirement of Article III; the Supreme Court reversed. A plaintiff need not breach an agreement and expose itself to suit in order to challenge the agreement’s validity in federal court. The case confirms that a remanufacturer or other industry plaintiff can file the patent-bargain case from the safe side of the regulatory wall, without first manufacturing a vehicle in defiance of federal regulation.

Nollan v. California Coastal Commission (1987; covered in the filability and objections-considered chapters). The Nollans owned a beachfront lot in Ventura County and applied for a permit to replace their bungalow with a larger house. The California Coastal Commission granted the permit on condition that they dedicate a public easement across their beach. The Supreme Court held that a regulatory condition must bear an essential nexus to the harm the regulation claims to address; an easement allowing the public to walk along the beach had no such nexus to the visual obstruction the larger house was said to cause. The case is the first prong of the modern proportionality test for regulatory conditions on property use, completed in *Dolan v. City of Tigard* (1994) and reaffirmed in *Sheetz v. County of El Dorado* (2024).

Patterson v. Kentucky (1878; covered in the filability and objections-considered chapters). The State of Kentucky banned the sale of a patented “improved burning oil” that the state inspector had condemned as unsafe. The patentee argued that his federal patent placed the product beyond state regulatory reach. The Supreme Court rejected the argument unanimously. The patent secured the inventor’s exclusive right to manufacture and sell against private competitors; it did not exempt the tangible product from state police-power regulation aimed at the safety of the public. The case is the foundational holding that a patent confers no immunity from generally applicable regulation — a holding the book accepts and builds on. The remanufacturer of a patent-expired design asks for the same regulation, not exemption from it.

Penn Central Transportation Co. v. City of New York (1978; covered in the two constitutional prong chapters). New York City’s landmarks law blocked the owner of Grand Central Terminal from constructing an office tower above the station. The owner claimed the restriction took its property. The Supreme Court ruled for the city and, in doing so, set out the framework that still governs most regulatory-takings claims: courts weigh the economic impact of the regulation on the owner, its interference with reasonable investment-backed expectations, and the character of the government action. Penn Central is the balancing test for regulations that diminish a property’s value without destroying it.

Pennsylvania Coal Co. v. Mahon (1922; covered in the two constitutional prong chapters). A Pennsylvania statute forbade underground coal mining that would cause the surface above to collapse, even where the coal company had sold the surface rights while expressly keeping the right to mine the coal below. Justice Holmes, writing for the Court, held that the statute went too far and effected a taking. The decision is the origin of regulatory-takings law: it established that although property may be regulated, a regulation that goes too far will be recognized as a taking requiring compensation.

Printz v. United States (1997; covered in the filability and objections-considered chapters). The Brady Handgun Violence Prevention Act required county sheriffs and other local chief law-enforcement officers to perform federal background checks on prospective handgun purchasers. Sheriff Jay Printz and Sheriff Richard Mack sued, arguing that the federal

government could not compel state officers to enforce federal regulatory programs. The Supreme Court agreed. The federal government may not commandeer the legislative or executive officers of the states for federal regulatory ends; doing so violates the structure of dual sovereignty the Constitution establishes. The case does not address “limited Times,” but it is the constitutional ground for the sheriff plaintiff — the elected county sheriff suing as a constitutional officer to challenge a federal regulatory program that converts him into an enforcement adjunct.

Romer v. Evans (1996; covered in the filability and objections-considered chapters). Colorado voters adopted a constitutional amendment forbidding the state or any locality to extend antidiscrimination protections on the basis of sexual orientation. The Supreme Court struck the amendment down on rational-basis review, holding that the only purpose the amendment could rationally be said to serve was disqualification of an unpopular group from political participation, which is not a legitimate state interest. The case extends the *Cleburne* line of “rational basis with bite” — the narrow set of cases in which an equal-protection challenge succeeds because the only rationale a court can find is identifiable animus.

Ruckelshaus v. Monsanto Co. (1984; covered in the two constitutional prong chapters). Monsanto challenged a federal pesticide statute that allowed the EPA to use and disclose the confidential test data that companies submitted for registration. The Supreme Court held that a trade secret is property protected by the Takings Clause, even though it has no physical form, and that a regulation appropriating the value of that confidential data can be a taking — though whether it is depends on whether the company held a reasonable investment-backed expectation that the data would stay confidential. The case establishes that intangible rights of defined economic value are property for Fifth Amendment purposes.

Scott Paper Co. v. Marcalus Manufacturing Co. (1945; covered in the two constitutional prong chapters). Marcalus had assigned a patent to Scott Paper, then later built a product using a design covered by a different patent that had already expired. Scott Paper invoked a contract doctrine — assignor estoppel — to stop its former assignor from competing. The

Supreme Court refused. The public’s right to use an invention whose patent has expired, the Court held, is a matter of patent policy that a private estoppel rule cannot override. The patent laws, the Court held, “dedicate to public use the invention of an expired patent,” and do not contemplate that anyone may “withhold from the public the use of the invention of an expired patent.”

Sears, Roebuck & Co. v. Stiffel Co. (1964; covered in the two constitutional prong chapters). Stiffel made a pole lamp, held patents on it that were later found invalid, and sued Sears for selling a near-identical copy under state unfair-competition law. The Supreme Court held that a State may not forbid the copying of an article that federal patent law leaves unprotected. What is unpatented or patent-expired is free for all to copy, and state law cannot re-create the monopoly federal law withholds. In the Court’s words, an article on which the patent has expired “may be made and sold by whoever chooses to do so.”

Sheetz v. County of El Dorado (2024; covered in the two constitutional prong chapters). A county charged a landowner a substantial traffic-impact fee as a condition of a building permit. The fee was fixed by a general schedule the county’s legislature had adopted, not by an official’s individual decision. The landowner raised a takings challenge; the county argued that the Takings Clause’s limits on permit conditions apply only to individualized administrative demands, not to conditions set by general legislation. The Supreme Court unanimously rejected the distinction: the Takings Clause contains no exception for legislatively imposed conditions.

Susan B. Anthony List v. Driehaus (2014; covered in the filability and objections-considered chapters). The Susan B. Anthony List, an antiabortion advocacy organization, had distributed political messages criticizing a congressional candidate for his health-care vote. The candidate filed a complaint under an Ohio statute criminalizing false statements in political campaigns. The advocacy organization sued in federal court before any prosecution had occurred, seeking a declaration that the statute violated the First Amendment. The Supreme Court held that pre-enforcement standing existed: a credible threat of enforcement under a statute the plaintiff intends to violate suffices to establish

injury in fact, and the plaintiff need not wait for prosecution to challenge the statute. The case is the controlling modern statement of pre-enforcement standing, and the doctrine on which the patent-bargain case will rest at the standing stage.

Tyler v. Hennepin County, Minnesota (2023; covered in the two constitutional prong chapters). A county seized an elderly homeowner’s condominium over a modest unpaid property-tax debt, sold it, and kept the entire sale price — including the surplus far exceeding what was owed. The county argued that state law had extinguished the owner’s claim to the surplus. The Supreme Court held that keeping the surplus was a taking. A government cannot sidestep the Takings Clause, the Court ruled, by enacting a law that redefines a long-recognized property right as no property at all.

Webber v. Virginia (1880; covered in the filability and objections-considered chapters). Two years after *Patterson v. Kentucky*, the Supreme Court reaffirmed the *Patterson* principle in the tax setting. Virginia imposed a license tax on the sale of sewing machines, and a patentee argued that his federal patent exempted the patented machines from the tax. The Court rejected the argument, holding that letters patent granted by the United States do not exclude the tangible property in which the invention is embodied from state tax or license law. The case is part of the *Patterson* line establishing that the existence of a patent does not place the patented product beyond ordinary state and federal regulation.

West Virginia v. Environmental Protection Agency (2022; covered in the two constitutional prong chapters). Several States challenged the EPA’s authority to reshape the nation’s electricity generation through climate regulation. The Supreme Court ruled against the agency and articulated the “major questions doctrine”: when an agency claims the power to decide a matter of vast economic and political significance, it must point to clear authorization from Congress, not infer the power from a vague or ancillary statutory phrase.

Wheaton v. Peters (1834; covered in the filability and objections-considered chapters). Henry Wheaton had been the Supreme Court’s third official reporter; Richard Peters succeeded him and began publishing condensed digests of the same opinions Wheaton had collected. Wheaton sued, claiming a common-law copyright in the published case reports.

The Supreme Court rejected the claim, holding that there is no common-law copyright in published works; intellectual-property rights exist only as Congress grants them under the Patent and Copyright Clause. The implication for “limited Times” is structural: the Patent Clause is the only source of patent rights, and a federal regulation that extends to a manufacturer a patent-equivalent commercial protection without the patent’s expiration cannot be defended on the theory that the protection is something other than a patent right.

Wickard v. Filburn (1942; covered in the chapter on the Patent Clause prong). Roscoe Filburn grew wheat on his Ohio farm in excess of his federal allotment, intending the surplus for his own use, not for sale. The Supreme Court upheld the penalty against him, reasoning that even purely local activity may be regulated under the Commerce Clause when its aggregate effect on interstate commerce is substantial. The case is the high-water mark of the modern commerce power and the authority a defender of the regulatory wall will cite for the proposition that Congress may regulate the manufacture of vehicles in commerce. The book does not dispute that power. The Patent Clause prong answers it on a different ground: a power broad enough to reach the wheat on Filburn’s farm is still not a power to cancel the public’s ownership of an expired invention, because no grant of federal power authorizes Congress to nullify what another clause of the Constitution commands.

Williamson v. Lee Optical of Oklahoma, Inc. (1955; covered in the filability and objections-considered chapters). Oklahoma required eyeglass replacement lenses to be made only from prescriptions written by licensed optometrists or ophthalmologists, against the challenge that the statute was economic protectionism dressed in health regulation. The Supreme Court upheld the statute unanimously, announcing what is now the rational-basis floor for federal economic classifications: a court asks only whether the classification is rationally related to some conceivable legitimate government interest, not whether the stated rationale is the real one. The case is the wall almost every Fifth Amendment equal-protection challenge to federal economic regulation runs into and breaks on, and it is the principal reason equality before the law cannot serve as a standalone constitutional prong of this book’s argument.

Appendix B: Note on AI-Assisted Authorship

On January 29, 2025, the U.S. Copyright Office published *Copyright and Artificial Intelligence, Part 2: Copyrightability* (copyright.gov/ai), concluding that works created by human authors using AI tools are eligible for copyright protection when the human provides substantial creative direction, selection, and arrangement. The Office confirmed that “the use of AI to assist in the process of creation or the inclusion of AI-generated material in a larger human-generated work does not bar copyrightability.” All constitutional arguments, litigation strategy, savings methodology, editorial structure, and substantive conclusions in this book are the original intellectual work of the author; AI tools served as research assistants and drafting aids under the author’s direction across multiple working sessions.

In April 2026, while this book was in its final review cycle, Jamie Metzl — technology futurist and author of six previous books — published *The AI Ten Commandments: A New Moral Code for Humanity* (Transcend Books) with GPT-5 named as his co-author, which Metzl described in *The Free Press* as “the first major work of nonfiction listing a human and AI as co-authors.” Metzl’s methodology runs closely parallel to the one used here. The difference in authorship credit is principled: Metzl’s book presents what an AI synthesized from the world’s moral traditions when given a careful prompt, so naming the AI as co-author is honest disclosure of what the book is. This book presents a constitutional argument no AI generated — the Patent Clause violation, the three-tier breakdown, the platform-year remedy, and the rest of the analytical spine came from a human reader spending three years across patent law, regulatory economics, and constitutional doctrine. The AI assisted in research, drafting, and refinement under substantial human direction, in the manner the Copyright Office’s 2025 guidance describes as eligible for full copyright protection.

Glossary

(Alphabetical by abbreviation. Acronyms are defined parenthetically at first appearance in the running text and collected here for reference.)

AEB — Automatic Emergency Braking. Federal Motor Vehicle Safety Standard requiring vehicles to detect imminent forward collisions with other vehicles or pedestrians and apply the brakes automatically when a driver does not respond. Codified by NHTSA in 2024 with phased compliance dates running through model year 2029.

AFBF — American Farm Bureau Federation. Largest U.S. general farm organization; publishes the Market Intel reports that track annual Chapter 12 farm bankruptcy filings drawn from U.S. Courts data and cited in the farm-equipment chapter.

BNCAP — Bharat New Car Assessment Programme. The vehicle safety assessment program of the Government of India, established in 2023 and operated under the Ministry of Road Transport and Highways. Tests new vehicles sold in India under crash protocols harmonized with international standards and awards adult occupant protection and child occupant protection ratings on a five-star scale.

CAFE — Corporate Average Fuel Economy. Federal fuel-economy standard administered by the National Highway Traffic Safety Administration; sets fleet-average fuel-efficiency targets that vehicle manufacturers must meet across their U.S. sales.

C.F.R. — Code of Federal Regulations. The codification of federal regulations published by the Office of the Federal Register and organized by title and section (e.g., 49 C.F.R. § 563 refers to Title 49, Section 563).

CJ — Civilian Jeep. The post-WWII civilian successor to the Willys MB military Jeep, manufactured under various model designations (CJ-2A, CJ-3A, CJ-3B, CJ-5, CJ-7) from 1945 through 1986.

CKD — Completely Knocked Down. A vehicle shipped as a kit of components, to be assembled in the destination country. The first Mahindra Jeep vehicles were assembled

from CKD kits shipped from the United States.

CPSC — Consumer Product Safety Commission. Independent federal agency established by the Consumer Product Safety Act of 1972; sets and enforces mandatory product-safety standards for consumer products under 15 U.S.C. §§ 2051 et seq.

DOE — Department of Energy. Cabinet-level federal department; sets and enforces appliance and equipment energy-efficiency standards under the Energy Policy and Conservation Act and its amendments.

DOJ — Department of Justice. Cabinet-level federal department; brings federal civil and criminal enforcement actions, including under the Federal Food, Drug, and Cosmetic Act and consumer-protection statutes.

EISA — Energy Independence and Security Act of 2007. Federal statute (Pub. L. 110-140) whose lighting-efficiency provisions amended the Energy Policy and Conservation Act and effectively withdrew the ordinary incandescent bulb from the U.S. residential lighting market over a phase-in period running from 2012 to 2023.

EPA — Environmental Protection Agency. Independent federal agency; sets and enforces air-quality, emissions, and environmental-compliance standards under the Clean Air Act and related statutes.

ESC — Electronic Stability Control. Federal Motor Vehicle Safety Standard No. 126; required of all light vehicles manufactured for sale in the United States since model year 2012. Detects loss of steering control and automatically applies individual wheel brakes to restore directional stability.

ESG — Environmental, Social, and Governance. A class of investment-screening criteria used by institutional capital allocators — banks, private-equity funds, pension funds, and other large pools of capital — to evaluate the environmental, social, and corporate-governance characteristics of companies and projects considered for investment.

EUA — Emergency Use Authorization. Authority delegated to the FDA Commissioner under the Federal Food, Drug, and Cosmetic Act to permit the use of unapproved medical

products, or unapproved uses of approved medical products, during a declared emergency.

FARFA — Farm and Ranch Freedom Alliance. Coalition-building organization founded in 2006 representing independent small-scale farmers, ranchers, and value-added producers.

FDA — U.S. Food and Drug Administration. Independent federal agency within the Department of Health and Human Services; regulates drugs, biologics, medical devices, food, cosmetics, and tobacco products under the Federal Food, Drug, and Cosmetic Act.

FMVSS — Federal Motor Vehicle Safety Standards. The body of federal vehicle-safety regulations administered by the National Highway Traffic Safety Administration and codified at 49 C.F.R. Part 571.

GNCAP — Global New Car Assessment Programme. The London-based international vehicle safety assessment program, founded in 2011 as a coordinating umbrella for regional NCAPs. Operates the #SaferCarsForIndia testing program and other regional programs, awarding adult and child occupant protection ratings on a five-star scale based on harmonized crash test protocols.

HCQ — Hydroxychloroquine. Antimalarial drug first patented in the 1950s and on the World Health Organization Essential Medicines List.

HVAC — Heating, Ventilation, and Air Conditioning. The equipment category covering residential and commercial systems for indoor climate control; one of the consumer-durable categories most heavily affected by Department of Energy efficiency standards under 42 U.S.C. § 6295(o)(1).

IAMERS — International Association of Medical Equipment Remarketers and Servicers. Industry association representing independent third-party servicers of medical devices.

IIJA — Infrastructure Investment and Jobs Act of 2021 (also styled the Bipartisan Infrastructure Law). Pub. L. 117-58. Federal statute directing the Department of Transportation to issue performance standards for crash avoidance technology, including automatic emergency braking and impaired-driving prevention systems.

IMO — International Maritime Organization. United Nations specialized agency responsible for the safety and security of shipping and the prevention of marine and atmospheric pollution by ships. Develops the SOLAS, MARPOL, and COLREG conventions implemented in U.S. law through the Coast Guard’s vessel inspection regime.

IMV — Innovative International Multi-purpose Vehicle. Toyota Motor Corporation’s body-on-frame platform underlying the Hilux pickup, Fortuner SUV, Innova MPV, and Hilux Champ models manufactured in Thailand, Indonesia, and other Asian markets. The platform is built on a long-mature mechanical design whose underlying patents have been in the public domain for many years.

ISO 9001 — International Organization for Standardization, quality management systems standard. Voluntary international standard certifying that a manufacturer maintains a documented quality management system; cited in the farm-equipment chapter as one of the certifications Indian tractor manufacturers meet on equipment exported to global markets.

IVM — Ivermectin. Antiparasitic drug discovered in the 1970s, on the World Health Organization Essential Medicines List, and recognized with the 2015 Nobel Prize in Physiology or Medicine for the underlying discovery.

LCD — Local Coverage Determination. A coverage policy issued by a Medicare Administrative Contractor under authority delegated by the Centers for Medicare and Medicaid Services; specifies whether a particular item or service is reasonable and necessary, and therefore covered under Medicare Part A or Part B.

LED — Light-Emitting Diode. The semiconductor lighting technology that, under the Department of Energy efficiency rules finalized in 2023, has effectively replaced the incandescent bulb in U.S. residential lighting.

LRA — Legislative Reorganization Act of 1970. Federal statute that restructured congressional procedure by mandating recorded floor votes, open committee proceedings, and other “sunshine” reforms.

MAP-21 — Moving Ahead for Progress in the 21st Century Act. Pub. L. 112-141 (July 6, 2012). Federal surface-transportation reauthorization statute under which Congress directed NHTSA to issue lighting and marking standards for agricultural equipment, codified at 49 CFR Part 562 in 2016.

MB — Methylene blue. Synthetic dye first prepared in 1876; FDA-approved for the treatment of methemoglobinemia and used off-label for a range of clinical and supplement purposes.

NAECA — National Appliance Energy Conservation Act of 1987. Public Law 100-12, signed by President Reagan on March 17, 1987. The Act amended the Energy Policy and Conservation Act of 1975 and added the anti-backsliding provision codified at 42 U.S.C. § 6295(o)(1) — the keystone of the regulatory ratchet.

NEMA — National Electrical Manufacturers Association. Trade association representing the major U.S. manufacturers of electrical equipment, including the patent-protected lighting incumbents (GE Lighting under Savant Systems, Signify, formerly Philips, Osram, Cree, and Acuity). NEMA's lighting members actively lobbied for the federal general service lamps efficiency standards finalized between 2022 and 2024 that withdrew the ordinary incandescent bulb from the U.S. residential market; the healthy-lighting chapter develops this dynamic as the bootleggers-and-Baptists configuration at the center of the case.

NHTSA — National Highway Traffic Safety Administration. Federal agency under the Department of Transportation; sets and enforces the Federal Motor Vehicle Safety Standards.

NMOG — Non-Methane Organic Gas. A class of regulated tailpipe emissions covering hydrocarbons other than methane; one of the principal pollutant categories in EPA's Tier 2 and Tier 3 motor-vehicle emission standards, where it is typically reported in milligrams per mile.

NO_x — Nitrogen Oxides. Reactive gases formed by combustion in the presence of nitrogen, regulated as a tailpipe emission under the Clean Air Act. The “x” is a variable

subscript: NO_x covers nitric oxide (NO) and nitrogen dioxide (NO₂).

NRDC — Natural Resources Defense Council. Nonprofit environmental advocacy organization; named respondent in *Chevron U.S.A., Inc. v. Natural Resources Defense Council, Inc.* (1984).

NSAC — National Sustainable Agriculture Coalition. Coalition of grassroots organizations advocating for sustainable and organic farmers; has published analyses documenting that four firms control 45 percent of global farm equipment sales.

NTTAA — National Technology Transfer and Advancement Act of 1995. Pub. L. 104-113. Federal statute directing federal agencies to use technical standards developed by private-sector voluntary consensus standards bodies in lieu of developing federal standards from scratch.

OEM — Original Equipment Manufacturer. The producer of a finished product (medical device, vehicle, appliance, or other equipment); often used in contrast with third-party servicers and remanufacturers.

OSH Act — Occupational Safety and Health Act of 1970. Pub. L. 91-596. Federal statute establishing workplace safety standards through the Occupational Safety and Health Administration; codified at 29 U.S.C. §§ 651 et seq.

OSHA — Occupational Safety and Health Administration. Federal agency within the U.S. Department of Labor; codifies workplace safety standards including 29 CFR 1928.51, the 1976 rollover rule on agricultural tractors discussed in the farm-equipment chapter as the first of the three regulatory layers.

PRP — Platelet-Rich Plasma. A preparation derived from a patient's own blood by centrifugation, used for autologous orthopedic injection.

RCRA — Resource Conservation and Recovery Act of 1976. Pub. L. 94-580. Federal statute establishing the federal regulatory framework for solid and hazardous waste under 42 U.S.C. §§ 6901 et seq.

TPMS — Tire Pressure Monitoring System. Federal Motor Vehicle Safety Standard No. 138; required of all light vehicles manufactured for sale in the United States since September 1, 2007. Sensors detect significantly under-inflated tires and warn the driver.

TRIPS — Agreement on Trade-Related Aspects of Intellectual Property Rights. Annex 1C of the 1994 Marrakesh Agreement Establishing the World Trade Organization, binding on all 166 WTO member nations. Article 33 sets the minimum patent term at twenty years from the filing date; the post-expiration commons it implies is the international counterpart to the U.S. Constitution’s “limited Times” requirement.

U.S.C. — United States Code. The codification of the general and permanent laws of the United States, organized by title and section (e.g., 42 U.S.C. § 6295(o)(1) refers to Title 42, Section 6295, subsection (o)(1)).

USD — United States Dollar. The currency of the United States; used throughout the book in international price comparisons where foreign-currency prices are converted to U.S. dollar equivalents for the American reader.

USDA — U.S. Department of Agriculture. Cabinet-level federal department; the Economic Research Service publishes the Food Dollar Series cited in the farm-equipment chapter, breaking each U.S. retail food dollar into the farm share (15.9 cents in 2023) and the marketing share (84.1 cents).

USP — United States Pharmacopeia. The non-governmental standards-setting body whose monographs for drug substances and dosage forms are incorporated by reference into FDA regulation under the Federal Food, Drug, and Cosmetic Act.

WTO — World Trade Organization. Intergovernmental organization administering the global trade-rules framework, including the TRIPS Agreement under which all 166 member nations are bound by a minimum twenty-year patent term — the international counterpart to the U.S. Constitution’s “limited Times” requirement.

Key Terms

(Alphabetical by term. The book's specific terms of art, with brief definitions of how the book uses each.)

Anti-backsliding provision. Section 6295(o)(1) of Title 42 of the United States Code, added by the National Appliance Energy Conservation Act of 1987. The statute provides that an energy-efficiency standard set by the Department of Energy can be tightened over time but never relaxed. The book treats this as the keystone of the regulatory ratchet: the legislative mechanism that converts a temporary regulatory burden into a permanent one and, by extension, converts the lengthening of an expired patent's effective monopoly into a permanent constitutional violation.

Blockade pattern. The book's term for the more common form the regulatory ratchet takes: the federal regulatory wall is built around a technology before the patents covering it expire, so that by the time the patents expire the public's right to remanufacture comes into existence inside a regulatory environment that already forbids it. The 2005 Ford F-150 case is a blockade-pattern case. Most expired-patent commerce in the United States is blockade-pattern. The injury is harder for an ordinary American to see because the manufacturing window the public was entitled to never opened. The economic consequences are large; the intuitive force of the injury is muted by the absence's invisibility.

Bootleggers-and-Baptists. A coalition framework first named by the economist Bruce Yandle (1983) to describe regulatory regimes supported simultaneously by a moralizing public-interest constituency (the "Baptists," who believe in the regulation's stated purpose) and a profit-taking commercial constituency (the "bootleggers," who profit from the regulation's suppression of competition). The book argues that the post-1970 federal regulatory state is the bootleggers-and-Baptists architecture writ very large, with each leg of the regulatory wall genuinely supported by both wings of its coalition, which is what makes the coalition durable.

Combinatorial property of the public domain. The book's term for the engineering fact that an expired patent does not enter the public domain alone. It enters as a component,

available to be combined with every other expired-patent component, in any combination an engineer can devise. The remanufactured American product that emerges when the regulatory wall is lifted is therefore not a copy of any single expired-patent design but a freshly engineered assembly of public-domain parts. The public domain is a kit of parts, not a museum.

Constitutional forgetting. The book’s term for the operational disappearance of the words “limited Times” from the Patent Clause as applied by Congress, the federal regulatory agencies, and the academic disciplines that study them. The verb is “forgetting” — not “misreading” or “ignoring” — because the doctrine was never overturned. Supreme Court precedent on the post-expiration public domain remains intact and unbroken; what has been lost is the operational memory of the bargain in the institutions whose work bears on it.

Expired-patent commerce. The book’s denominator term for the share of American economic activity in which the underlying patents have expired. The book divides this commerce into three tiers: 58 percent Blocked (where federal regulation makes remanufacturing legally impossible), 15 percent Hindered (where federal regulation imposes substantial but non-prohibitive constraints), and 27 percent Flowing (where remanufacturing operates freely). The relevant denominator is expired-patent commerce, not “the economy” — which is why the 58 percent figure does not refer to all American manufacturing.

Expired Patent Commons. The book’s term for the destination an invention is meant to reach when its patent expires: the body of expired inventions that the public may freely make, copy, improve, and remanufacture. The word “commons” is load-bearing — it denotes a shared resource the public can actually reach and use, not merely a right that has lapsed on paper. An invention has entered the Expired Patent Commons only when an American may in fact build it; a patent that has expired but that federal regulation forbids anyone to manufacture has not. The Expired Patent Commons is the public’s half of the patent bargain, and the book’s central claim is that the post-1970 regulatory wall prevents inventions from entering it — the forgotten half of the Patent Clause. The Supreme Court’s cases call the same destination the public domain in the patent-law sense.

Grandfather exemption. The first leg of the book’s three-leg remedy. A judicial or legislative carve-out under which a remanufactured product built on a public-domain platform is held to the federal safety and energy-efficiency standards in effect during the platform year the remanufactured product mimics — the same standards the original product was certified to. The grandfather exemption flows naturally from a successful constitutional ruling on the Patent Clause and requires no act of Congress to put in place.

Health-freedom statute. A state-level statute that limits the authority of state medical or pharmacy licensing boards to discipline licensed physicians for the use of treatments not approved by the U.S. Food and Drug Administration, provided the practitioner has obtained informed consent and represents the modality accurately. Roughly fifteen states have enacted such statutes since the late 1990s. Relevant to the Hindered sector, where licensing-board discipline is a common enforcement angle.

Hindered sector. The roughly fifteen percent of expired-patent commerce — chiefly pharmaceuticals and medical devices — where federal regulation, professional-board enforcement, and insurance-coverage decisions impose substantial but non-prohibitive constraints on use of off-patent products. The Hindered sector contrasts with the Blocked sector, in which post-1970 federal manufacturing mandates make remanufacturing legally impossible, and with the Flowing sector, in which remanufacturing operates freely.

Limited Times. The two words at the heart of Article I, Section 8, Clause 8 of the U.S. Constitution: “To promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries.” The phrase is constitutive, not decorative — affirmed across more than thirty-eight Supreme Court holdings spanning two centuries. The book’s central claim is that the post-1970 regulatory state has converted “limited Times” into perpetual times across roughly 58 percent of expired-patent commerce, in violation of the Patent Clause.

Median. A statistical measure of central tendency: the middle value of a distribution, where half of cases fall above and half fall below. The median is distinct from the mean (or arithmetic average), which is the sum of all values divided by the count. The book uses

median household income — \$83,730 in 2024, per the U.S. Census Bureau — rather than mean household income because the right-skewed distribution of American household income (a small number of very high earners pulling the mean upward) makes the median the better description of where the typical American household actually earns. The same logic applies to median wage, median home price, and other median figures the book cites.

Public domain (patent-law sense). The constitutional status of an invention after its patent has expired. In the patent-law sense (distinct from the colloquial sense, in which “public domain” refers loosely to anything freely available), the public domain is the operational result of the bargain the Patent Clause establishes: the inventor receives an exclusive right for a limited time, after which the invention belongs to the American public to make, copy, improve, and remanufacture. The Supreme Court has repeatedly affirmed that the post-expiration public domain is a vested public right.

Regulatory ratchet. The book’s term for the one-way mechanism by which federal regulation tightens over time but cannot be relaxed, with the result that the original legal terrain — including, crucially, the patent commons that emerges when patents expire — becomes progressively narrower. The ratchet is uniquely hostile to the Patent Clause because it converts a temporary regulatory burden into a permanent extension of monopoly protection. The anti-backsliding provision (42 U.S.C. § 6295(o)(1)) is the keystone statutory expression of the ratchet.

Remanufacturing (book’s specific sense). The act of building a freshly engineered product assembled from components whose individual patents have all expired. The book’s sense is distinct from “restoration” (returning a specific used object to working condition) and from “reproduction” (manufacturing a copy of a specific older design). A remanufactured product is a kit-of-parts assembly: a chassis whose patents expired in the early 2000s, a USB port whose 1995 Intel patents expired around 2015, a backup camera whose imaging patents expired years ago — engineered, mixed, and matched to deliver the features buyers actually want at a fraction of the new-vehicle price. The public domain is a kit of parts, not a museum.

Sunset Amendment. The book’s term for the proposed constitutional amendment introduced in the chapter on the environmental question. The amendment preserves the Patent Clause remedy at full strength for twenty years from the date of first lawful remanufacture in any product category, after which Congress may by law restrict or prohibit further remanufacture without obligation to provide just compensation under the Fifth Amendment, applying only prospectively. The amendment is offered as Phase 2 structural insurance: not required for the Phase 1 remedy to operate, but available if the country eventually decides to close the marginal compensation subsidy created by the Takings Clause backstop.

Three-tier breakdown. The book’s category-level audit of expired-patent commerce, dividing it into three tiers: Blocked (58 percent — sectors where federal manufacturing mandates make remanufacturing legally impossible, including vehicles, agricultural and construction machinery, HVAC, appliances, and certain lighting), Hindered (15 percent — sectors where federal regulation imposes substantial but non-prohibitive constraints, principally pharmaceuticals and medical devices), and Flowing (27 percent — sectors where remanufacturing operates freely, including bicycles, hand tools, firearms, and most consumer electronics). The breakdown is built from BLS Consumer Expenditure Survey data, BEA Industry Economic Accounts, and Census Bureau Annual Survey of Manufactures data; the methodology is detailed in the published companion methodology article — see the Endnotes entry for Martin, “The Forgotten Half of the Patent Clause: What was Promised When Patents Expired” (April 10, 2026).

Two suspenders. The book’s metaphor for its two independent constitutional prongs, either of which is sufficient on its own. The first prong is the Patent Clause: Article I, Section 8, Clause 8 authorizes Congress to grant exclusive rights only “for limited Times,” and a regulatory scheme that converts those limited times into perpetual protection exceeds the power the Clause actually grants. The second prong is the Takings Clause: the public’s right to copy and remanufacture an expired patent is a vested right, affirmed in *Bonito Boats* (1989), and a permanent regulatory cancellation of that right is a taking requiring just compensation. A litigant who loses on one prong can still win on the other.

Withdrawal pattern. The book’s term for the sharper second form the regulatory ratchet takes: the technology was already in the public domain, the public was already exercising the right to manufacture, and federal regulation was then erected around the technology after the fact, withdrawing it from the domain it had already been part of. The ordinary incandescent light bulb is the canonical case — the foundational patents expired more than a century ago, Americans manufactured incandescent bulbs continuously from 1879 onward, and federal energy-efficiency rules issued between 2007 and 2023 progressively withdrew the bulb from manufacturability. The withdrawal pattern is doctrinally stronger than the blockade pattern because the right at issue was not theoretical — it had been exercised in commerce, in plain view, for generations.

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— U.S. Constitution, Article I, Section 8, Clause 8